



Reproductive and metabolic toxic effects of polystyrene microplastics in adult female Wistar rats: a mechanistic study

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Abstract

Microplastics, such as polystyrene microplastics (PS-MPs), have become an emerging environmental hazard for animals and humans. Long-term exposure to PS-MPs has led to neurotoxicity, reproductive dysfunction, and carcinogenesis. The goal of this study was to evaluate the effect of sub-chronic exposure of PS-MPs on metabolic and reproductive functions in female rats. The PS-MPs were prepared by cryogenic technique. The PS-MPs were given orally to female Wistar rats for 45 days at 2.5, 5, and 10 mg/kg/day. The average PS-MPs' size diameter was 876 nm. The PS-MPs administration resulted in a significant decrease in the activity of superoxide dismutase and catalase in the liver and ovary. The effect of PS-MPs on reduced glutathione and lipid peroxidation in the liver and ovarian tissues of rats was statistically insignificant. The PS-MP exposure exhibited an increase in the levels of triglycerides, total cholesterol, and low-density lipoprotein and decrease in high-density lipoprotein. The PS-MPs caused glucose intolerance and increase in insulin. Moreover, the PS-MP exposure increased follicle stimulating hormone, estradiol, and testosterone. Serum level of interleukin-6 and nuclear factor kappa B (NF-κB) was elevated in animals treated with PS-MPs. The PS-MP exposed rats showed normal ovarian histology, but activated hepatic stellate cells and liver fibrosis. It is concluded that the sub-chronic exposure to PS-MPs resulted in metabolic and endocrine disruption in female rats through oxidative damage, hormonal imbalance, and chronic inflammation.

Keywords Microplastic · Polystyrene · Endocrine disruption · Hepatotoxicity · Oxidative stress · Inflammation

Introduction

Synthetic polymers are the most important type of materials in the twenty-first century and used everywhere and have a significant impact on our daily lives (Lodge 2017). They are classified into thermoplastic, thermosetting, elastomers,

and synthetic fibers (Fried 2014). Polystyrene (PS) is a thermoplastic polymer used in food packaging industry (Piringer and Baner 2008). As manufacturing rises, so does the amount of plastic garbage produced and accumulated in the environment. Over 250,000 tons of plastic pieces have accumulated in water bodies due to a lack of proper waste management (Eriksen et al. 2014). The breakdown of macroplastics into microplastic (MPs), caused by photodegradation, oxidation, hydrolytic degradation, and mechanical disintegration yielding pieces smaller than 5 mm, is called secondary microplastics (Cole et al. 2011). These MPs, also deliberately produced by industries to be used in cosmetics such as toothpastes or exfoliants, are termed as primary microplastic (Browne et al. 2011). An examination of tap water samples indicated a large volume of drinking water around the world contaminated with microscopic fragments (Guzzetti et al. 2018; Germoush et al. 2022; Manikkam et al. 2013). Microplastic (MP < 5 mm) debris adversely affects marine and freshwater organisms. These nano-sized, nanoplastic (NP ≤ 100 nm) polymers interact with living organisms and pass through biological

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membranes, penetrate into tissues, and accumulate in organs resulting in systemic exposure (Amereh et al. 2020). Humans are exposed to plastic particles through diet, cutaneous contact, and inhalation. When MPs build up in organisms, these exhibit a variety of negative consequences, including decreased feeding activity, mortality, stunted growth and development, endocrine disruption, oxidative stress, energy disruption, immune and neurotransmission failure, and even genotoxicity usually through generation of reactive oxygen species (ROS), oxidative stress, and lowering the immune cells' phagocytic activity (Bin-Jumah et al. 2021; Kabir et al. 2021; Lu et al. 2016). The styrene used in the manufacturing of PS has long been suspected of being an endocrine disruptor (Amereh et al. 2020).

The major pathways to the exposure of nanoplastics are through the lungs, gastrointestinal tract, and skin (Kosuth et al. 2018). One of the major routes of human exposure to MPs is through ingestion, and the estimated consumption of MPs through foodstuff is 39,000 to 52,000 particles per individual per annum (Cox et al. 2019). These MPs have been found in mussels, commercial fish, table salt, and bottled water (Li et al. 2016). With a surface area of 200 m², the gastrointestinal tract is the principal exposure location for the absorption of MPs (Pitt et al. 2018). Particles could be adsorbed in the intestine by specific M-cells that cover lymphoid tissue in the gut — Peyer's patches after ingestion, with the particle clearance rate dependent on adhesion to gastrointestinal mucus. The small-sized particles may enter the intestinal mucus due to adsorption (Ensign et al. 2012). The *in vitro* studies have shown that PS microparticles (PS-MPs) internalize and translocate in intestinal cell monocultures, with activation of interleukin (IL)-6 and IL-8 genes involved in stomach diseases (Forte et al. 2016). Another proposed mechanism of absorption for MPs is through paracellular pathway. Cellular uptake of small-sized particles is affected by interaction with biological elements such as proteins and phospholipids. Protein adsorption occurs from the bodily fluids on the surface of nanoplastics causing a "protein corona." The PS-MPs also form protein corona complexes resulting in enhanced translocation, cellular uptake, and toxicity (Lehner et al. 2019). The cellular internalization of PS-MPs is accomplished through a clathrin-independent absorption process (Rossi et al. 2014). The uptake of nanoplastics and MPs occurs mainly through endocytosis as well as through cell-specific pathways and passive diffusion mechanism (Wang et al. 2012).

Depending on the susceptibility and exposure to organisms, the MPs are now considered as hazardous. The MPs can cause oxidative stress, translocation to other tissues, and cytolysis, yet they are difficult to eliminate from the body, resulting in chronic inflammation. Increased exposure to MPs in the environment may potentially be contributing to the increased incidence of neurological diseases, cancer,

and immunological problems (Crawford and Quinn 2017). The key mechanism of MP-induced toxicity is oxidative stress and cytotoxicity. Oxidative stress associated with MPs occurs due to the high surface area and generation of oxidizing species, and the ROS released during the inflammatory response as previously shown in zebra fish and mice exposed to MPs (Prata et al. 2020); (Lu et al. 2016). Cytotoxicity is caused by MPs through inflammation, oxidative stress, and particle toxicity. PS-MPs are internalized into macrophages, erythrocytes, and rat alveolar cells (Geiser et al. 2005; Yacobi et al. 2008). The MPs are internalized by cells and interact with intracellular components. Furthermore, the PS of 60 micron size produced ROS and endoplasmic reticulum stress in lung macrophages and epithelial cells leading to autophagy (Chiu et al. 2015). The MPs are translocated to distant tissues of rats, the liver or spleen following inhalation and ingestion (West-Eberhard 2019).

In recent studies, the reproductive toxicity of PS-MPs was also evident in male rats (Amereh et al. 2020). Infertility is a stressful condition that adversely affects the mental health. It causes mental distress, disappointment, and a sense of loneliness (Amini et al. 2020). The main cause of female infertility is polycystic ovary syndrome (PCOS), accounting for 46.6% of all infertility cases. PCOS equally affects lean and obese women. Infectious causes of tubal factor infertility include pelvic inflammatory illness and tuberculosis. The second most prevalent cause of infertility is tubal disease, which affects 33.8% of females. Endocrine abnormalities and uterine fibroids associated with environmental causes are responsible for almost 20% of female infertility (Deshpande and Gupta 2019).

The PS persists in the environment and is converted to nanoplastics and MPs that are taken up by the cells and concentrated in various body tissues. Therefore, these PS-MPs may have endocrine disruptive effect in female animals possibly through oxidative stress, lipid peroxidation, insulin resistance, hyperleptinemia, hyperinsulinemia, and hyperandrogenism. This research work was aimed to evaluate the toxic effect of PS-MPs exposure on the liver and reproductive system of female rats on sub-chronic exposure.

Materials and methods

Chemicals

The polystyrene (PS) plastic (MS-Polymer, Pakistan), estradiol valerate (Hebei Mingyi Tech Co. China), and carboxymethyl cellulose (CMC), pyrogallol, thiobarbituric acid (TBA), diethyl ether, trichloroacetic acid (TCA), phosphate buffer solution (PBS), potassium phosphate (K₃PO₄), and dithiobis nitrobenzoic acid (DTNB) were obtained from Sigma-Aldrich® (Germany). Moreover, glucose (Unilever

foods limited, Pakistan), diagnostic kits for follicle stimulating hormone (FSH), luteinizing hormone (LH), estrogen (E2), progesterone, testosterone, lipid profile, nuclear factor erythroid 2-related factor 2 (Nrf-2), interleukin (IL-6), and nuclear factor kappa B (NF- κ B) assay kits (Elab Science, China) were also used in the study.

Preparation of PS-MPs

The cryogenic technique was used to prepare PS-MPs from 2- to 5-mm pellets of PS (Astner et al. 2019). The pellets were kept in a freezer at -18°C for 3 days which were then blended and fractionated by passing gradually through various sieves of mesh size 80, 120, and 400 for size separation. For particle size determination, a suspension of PS-MPs in water was prepared and evaluated using Zeta sizer (Malvern, England).

Experimental animals

Thirty female Wistar rats of weight ranging from 210 to 240 g and age of more than 12 weeks were acquired from Riphah Institute of Pharmaceutical Sciences and divided into five equal groups. All animals were housed in the animal house of Riphah Institute of Pharmaceutical Sciences (RPS) at a temperature of $25 \pm 3^{\circ}\text{C}$ having 55–70% humidity. For 2 weeks, tap water and a standard pellet diet were given to animals to acclimatize with the laboratory conditions (Capdevila et al. 2007). The animal study was approved by the university Institutional Ethics committee, Reference number REC/RIPS-LHR/2022/518.

Experimental design

Animals in group I were designated as normal control which received standard pellet diet and water only. Animals in group II (disease control group, i.e., CEV) received estradiol valerate (0.8 mg/kg once peroral) and 1 ml of 0.5% CMC solution only with a gavage (Lokind et al. 1991); (Mirabolghasemi and Kamyab 2017). Animals in groups III, IV, and V received PS-MPs 2.5, 5, and 10 mg/kg/day in 0.5% CMC suspension for 45 days according to previous studies to demonstrate the sub-chronic toxic effect of PS-MPs (Wei et al. 2022). Dosage of PS-MPs was selected based on previous investigations which studied environmentally relevant concentrations in rodents (Jin et al. 2021); (Ijaz et al. 2021); (Park et al. 2020).

After 45 days of dosing with PS-MPs, all rats were given anesthesia with ether, and their blood samples were taken by heart puncture. Serum samples were collected by centrifugation (centrifuge by Centurion scientific, UK) at 2500 rpm for 15 min and then stored at -4°C until analysis.

Biochemical analysis

For oral glucose tolerance test (OGTT), the rats were given 2 g glucose /kg body weight via oral gavage in a 40% aqueous solution following 45 days of treatment with PS-MPs. Blood samples were obtained from the tail vein at 60, 120, and 180 min following oral administration of glucose (Zulfqar et al. 2020). The concentrations of estradiol, LH, FSH, progesterone, and testosterone in the blood samples were analyzed. Moreover, serum levels of NF- κ B, IL-6, and Nrf-2 were determined with enzyme-linked immunosorbent assay (ELISA) kits according to manufacturer protocols (Hussain et al. 2016). Serum insulin, blood glucose, and lipid profile parameters such as triglycerides (TG), low-density lipoproteins (LDL), high-density lipoproteins (HDL), and total cholesterol (TC) were estimated by standard methods (Mansouri et al. 2011).

Determination of oxidative stress

Ovaries and livers from animals were separated from the fallopian tube and visceral fats respectively and weighed. Organs of all groups were individually kept in ice cold PBS solution and kept in freezer for further study. Organs were individually homogenized with tissue homogenizer. Tissue homogenates were mixed with 0.1 M PBS (pH 7.4) to prepare 10% w/v tissue preparation. Afterwards, the tissue samples were centrifuged for 10 min at 2000 rpm to remove supernatant which was used to assess catalase (CAT), malondialdehyde (MDA), superoxide dismutase (SOD), and reduced glutathione (GSH) (Kumar and Gupta 2002). The protein content of ovarian and liver tissues was estimated by Folin phenol method (Saleem et al. 2022).

Estimation of GSH

Ellman's approach was used to determine GSH levels. One milliliter of tissue supernatant was mixed with 1 mL of 10% TCA, followed by adding 4 mL PBS and 0.5 mL DTNB reagent. Absorbance was noticed at 412 nm within 15 min. The GSH concentration was measured as nM of GSH per mg of protein (Sharif et al. 2022).

Estimation of SOD

To perform this assay, 0.1 mL of tissue aliquot was poured into the test tube followed by the addition of 2.8 mL of PBS of pH 7.4 and 0.1 mL pyrogallol solution. Absorbance was recorded for 5 min at 325 nm after every 30 s against a blank with Shimadzu uv-1601, Japan (Iftikhar et al.). The amount of enzyme necessary to inhibit pyrogallol autoxidation by

50% under the assay conditions was 1 unit of SOD. A linear regression equation was employed to measure SOD activity (Velma and Tchounwou 2010).

Estimation of MDA

The TBA reagent was utilized to carry out this test. One milliliter of tissue supernatant was poured in a test tube with 3-mL TBA reagent. The mixture was then immersed in a hot water bath at 80 °C for 15 min before cooling over an ice bath. The sample solution was centrifuged for 10 min at 3500 rpm to withdraw the superficial layer, whose absorbance was recorded at 532 nm after cooling (Saleem et al. 2021). MDA content was expressed in $\mu\text{g}/\text{mg}$ of protein.

Estimation of CAT

In this assay, the CAT activity was measured using Aebi's technique (Akhtar et al. 2021a). Tissue aliquot (0.05 mL) and 1.95 mL of 50 mM phosphate buffer were (pH 7.0) were admixed. Afterwards, 1 mL of 30 mM hydrogen peroxide (H_2O_2) was added, and dissimilarities in absorbance were measured at 15-s interval for 30 s at 240 nm (Bhangale and Acharya 2016). The activity of CAT was determined using milli-molar extinction coefficient of H_2O_2 ($0.071 \text{ mmol cm}^{-1}$) and the activity was expressed as micro-moles of H_2O_2 oxidized/min/mg protein.

Histological analysis of ovary and liver

For histopathological study, samples of ovaries and liver tissues were placed in 10% formaldehyde until cutting and embedding in paraffin wax. Tissue sections of 5- μm thickness were prepared, mounted on glass plates and stained with hematoxylin–eosin dye for histological observations under compound microscope (Akhtar et al. 2021b).

Analytical statistics

The results were presented in the form of mean $\pm$ standard deviation (S.D.). GraphPad prism 5 software was used to conduct statistical analyses using the one-way analysis of variance (ANOVA) test followed by Bonferroni's post hoc test. The results were considered moderately significant when $p < 0.01$ while they were considered extremely significant at $p < 0.001$ and $p < 0.05$.

Results

This study evaluated the harmful effects of PS-MPs on the metabolic and reproductive system in female Wistar rats. The characteristics of PS-MPs and their effect on the

oxidative stress in the liver and ovarian tissues, lipid profile, blood glucose, insulin, reproductive hormones, and histological parameters were determined.

Characteristics of PS-MPs

The viscosity and refractive index (RI) of the dispersant were 1.2 mPa S and 1.330 respectively. The RI of PS-MPs was 1.58 while the absorption of material was 0.010. The sample was placed in a glass cuvette for a count rate of 145.9 kcps and run for 40 S at 25 °C. The particle size obtained after sieve fractionation was $\leq 1 \mu\text{m}$, and the average particle size diameter was 876 nm.

Body and ovary weights

Administration of PS-MP at different doses resulted in significant changes in body weight of female Wistar rats following 30 and 45 days of exposure. However, treatment of animals with PS-MP at 10 mg/kg resulted in statistically significant reduction in ovary weight as compared to normal rats. The effect of PS-MPs exposure on ovary and body weights female Wistar rats is shown in Table 1.

Oxidative stress in the liver

When compared to normal control rats, the protein level in liver was considerably higher in CEV. The level of protein was not significantly modified in P 2.5 mg/kg but was significantly increased in P 5 and in P 10 mg/kg/day groups as compared to normal control group. The effect of microplastic exposure on oxidative stress parameters in the liver tissue of normal rats is shown in Fig. 1.

CAT activity

In the liver, CAT activity was lower in CEV ($2.36 \pm 0.16 \text{ U}/\text{mg}$ protein) compared to normal control group ($5.44 \pm 0.14 \text{ U}/\text{mg}$ protein). The activity of enzyme was significantly decreased in all PS-MP-treated groups (P 2.5 mg/kg: $3.91 \pm 0.12 \text{ U}/\text{mg}$ protein, P 5 mg/kg: $3.23 \pm 0.19 \text{ U}/\text{mg}$ protein, P 10 mg/kg: $2.68 \pm 0.22 \text{ U}/\text{mg}$ protein) in comparison to normal control group. It was also found that the activity of CAT was significantly higher in rats treated with 2.5 and 5 mg/kg in comparison CEV. The effect of PS-MPs exposure on the liver tissue CAT activity of female rats is shown in Fig. 1a.

GSH content

In the liver tissue, the content of GSH was unaffected in CEV ($3.88 \pm 0.42 \mu\text{M}/\text{mg}$) and PS-MP 2.5 mg/kg ($4.81 \pm 0.44 \mu\text{M}/\text{mg}$) groups and was comparable to

Table 1 Effect of polystyrene microplastic (PS-MP) exposure on body and ovary weight of female Wistar rats

Weight	Normal control	Estrogen treated control	PS-MP 2.5 mg/kg	PS-MP 5 mg/kg	PS-MP 10 mg/kg
Initial B.W	227.75 ± 10.96	217.83 ± 3.95	225.65 ± 10.81	220.78 ± 15.24	229.22 ± 8.48
B.W. after 15 days	228.55 ± 7.77	186.65 ± 9.40****	217.58 ± 10.71###	232.16 ± 3.06####	226.66 ± 8.01####
B.W. after 30 days	255.52 ± 6.36	195.43 ± 7.07****	219.22 ± 12.72*****##	235.58 ± 5.06*****	227.91 ± 3.53*****###
B.W. after 45 days	274.15 ± 7.28	215.21 ± 7.07****	228.55 ± 4.94****	235.91 ± 6.92*****##	224.83 ± 8.72****
Ovary weight	0.1625 ± 0.015	0.1125 ± 0.015**	0.1425 ± 0.0263	0.125 ± 0.01291	0.1245 ± 0.01037*

Results were calculated as mean ± S.D ($n=6$). Data for body weights were assessed by using RM two-way ANOVA followed by Tukey's post hoc test. Ovary weights were assessed by one-way ANOVA followed by post hoc test

B.W. body weight

**** $p < 0.0001$; *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$ show considerably different in comparison to normal control group, whereas #### $p < 0.0001$; ### $p < 0.001$; ## $p < 0.01$ show considerably different in comparison to estrogen-treated group

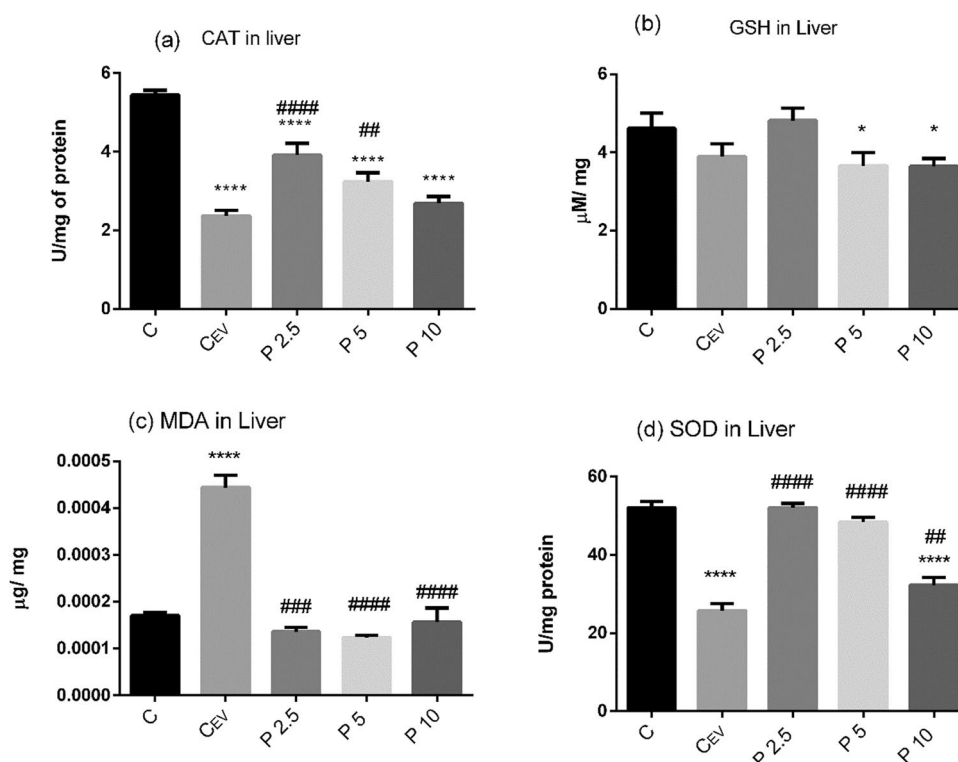


Fig. 1 Effect of polystyrene microplastic exposure on oxidative stress biomarkers in the liver tissue of female rats. **(a)** Catalase activity, **(b)** Reduced glutathione, **(c)** Malondialdehyde, and **(d)** Superoxide dismutase activity in the liver. Results were calculated as mean ± S.D ($n=6$). Data were assessed by using one-way ANOVA followed by Bonferroni's post hoc test. **** $p < 0.0001$ and * $p < 0.05$ showed considerably different in comparison to normal control group (C), whereas #### $p < 0.0001$, ### $p < 0.001$, and ## $p < 0.01$ showed con-

siderably different in comparison to estrogen-treated group (CEV). P 2.5: polystyrene microplastic 2.5 mg/kg/day, P 5: polystyrene microplastic 5 mg/kg/day, P 10: polystyrene microplastic 10 mg/kg/day. The PS-MP significantly reduced CAT activity in ovarian tissues at all dose levels. It significantly decreased the SOD at 10 mg/kg, while GSH at 5 and 10 mg/kg in liver tissues. The MDA level remained unaffected by PS-MP exposure

normal control group ($4.61 \pm 0.35 \mu\text{M/mg}$). The level of GSH in PS-MP 5 ($3.65 \pm 0.37 \mu\text{M/mg}$) and 10 mg/kg ($3.64 \pm 0.28 \mu\text{M/mg}$) groups was considerably reduced in comparison to normal control group. The effect of PS-MP exposure on GSH in liver tissue of female rats is shown in Fig. 1b.

MDA level

When CEV ($0.44 \pm 0.003 \text{ ng/mg}$) was compared to the normal control group ($0.17 \pm 0.022 \text{ ng/mg}$), the amount of MDA in the liver was only marginally affected. In PS-MP treated groups, the level of MDA was insignificantly increased as

dose of PS-MP increased (P 2.5 mg/kg: 0.13 ± 0.001 ng/mg, P 5 mg/kg: 0.12 ± 0.003 ng/mg, P 10 mg/kg: 0.15 ± 0.004 ng/mg). In comparison to normal control group, the MDA level was insignificantly different in all PS-MP treated rats. Moreover, PS-MP treated groups showed lower lipid peroxidation as compared to CEV. The effect of PS-MP exposure on MDA level in the liver of female rats is shown in Fig. 1c.

SOD activity

The SOD activity in the liver of CEV rats (25.70 ± 2.13 U/mg protein) was significantly lower as compared to the normal control group (52.05 ± 1.71 U/mg protein). Administration of PS-MP at 2.5 (52.13 ± 1.42 U/mg protein) and 5 mg/kg (48.34 ± 2.11 U/mg protein) to female rats resulted in indistinct effect on SOD as compared to normal control group. PS-MP 10 mg/kg showed significant reduction of SOD (32.29 ± 2.23 U/mg protein) in the liver as compared to the normal control group. Moreover, PS-MP at all doses showed statistically raised activity of SOD in comparison to CEV. The effect of PS-MP exposure on SOD activity in the liver of female rats is shown in Fig. 1d.

Oxidative stress in ovarian tissue

In normal rats, protein content in ovarian tissues was decreased in CEV as compared to normal control group. In all PS-MP treated groups, protein level was considerably reduced in comparison to normal control and CEV. The effect of PS-MP exposure on ovarian tissue oxidative stress parameters in female rats is shown in Fig. 2.

CAT activity

The effect of CAT activity in ovarian tissue was observed in Wistar rats treated with PS-MP in comparison with the normal control group (20.09 ± 2.01 U/mg protein) and CEV (11.11 ± 1.32 U/mg protein). Among PS-MP treated rats, the CAT activity was the highest in PS-MP 2.5 mg/kg. Treatment with PS-MPs at 5 and 10 mg/kg resulted in a significant decline in CAT activity in comparison to the normal control group. It was further found that CAT activity was significantly reduced in CEV as compared to the normal control group. It was also revealed that the CAT activity was significantly higher in PS-MP treated groups (P 2.5:

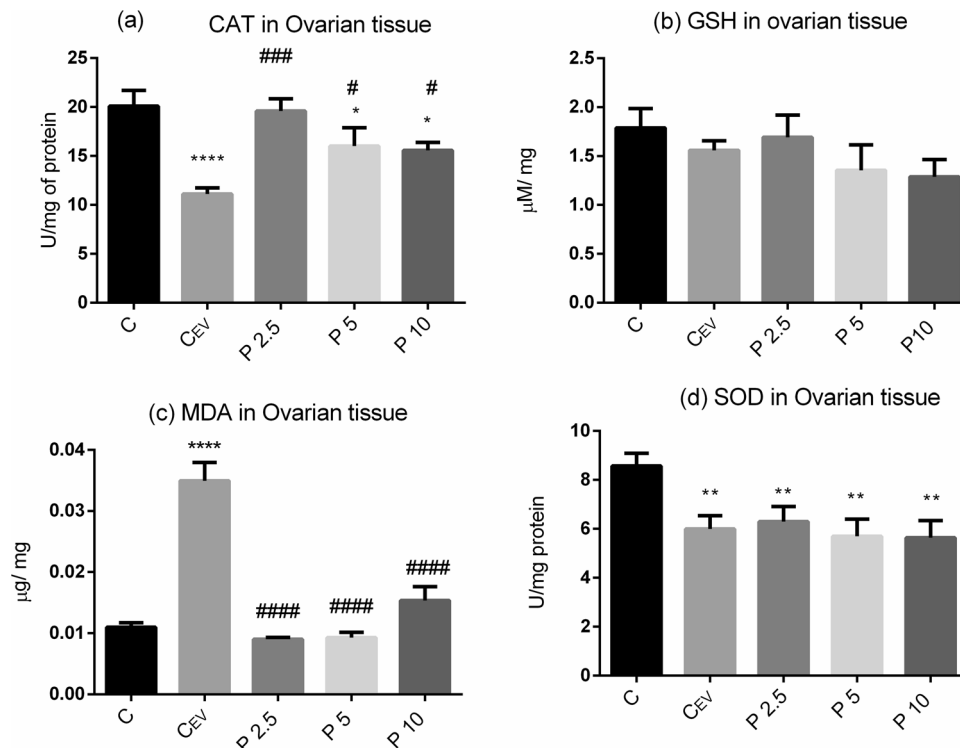


Fig. 2 Effect of Polystyrene microplastic exposure on oxidative stress biomarkers in the ovarian tissue of rats. **(a)** Catalase activity, **(b)** Reduced glutathione, **(c)** Malondialdehyde, and **(d)** Superoxide dismutase in the ovary. Results were calculated as mean $\pm$ S.D ($n=6$). Data were assessed by using one-way ANOVA followed by Bonferroni's post hoc test. **** $p < 0.0001$; *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$ showed considerably different in comparison to normal control group (C), whereas ##### $p < 0.0001$, ### $p < 0.001$; # $p < 0.05$

showed considerably different in comparison to estrogen-treated group (CEV). P 2.5: polystyrene microplastic 2.5 mg/kg/day, P 5: polystyrene microplastic 5 mg/kg/day, P 10: polystyrene microplastic 10 mg/kg/day. The effect of PS-MPs was insignificant on GSH and MDA while it significantly reduced the SOD in ovarian tissue. PS-MP also significantly reduced CAT activity in ovarian tissues at 5 and 10 mg/kg dose

19.59 ± 1.79 U/mg protein, P 5 mg/kg: 16.01 ± 1.98 U/mg protein, P 10 mg/kg: 15.56 ± 1.05 U/mg protein) groups as compared to CEV. The effect of PS-MP exposure on CAT activity in ovarian tissue of rats is shown in Fig. 2a.

GSH content

The amount of GSH in the ovarian tissue of CEV (1.55 ± 0.11 µM/mg) was only marginally reduced as compared to normal control group (1.78 ± 0.21 µM/mg). In PS-MP treated groups (P 2.5 mg/kg: 1.69 ± 0.22 µM/mg, P 5 mg/kg: 1.35 ± 0.33 µM/mg, P 10 mg/kg: 1.28 ± 0.23 µM/mg), the level of GSH was insignificantly different in comparison to CEV and normal control group. The effect of PS-MP exposure on GSH in ovarian tissue of Wistar rats is shown in Fig. 2b.

MDA level

In the ovary, MDA level was significantly lower in the normal control group (10.97 ± 0.51 ng/mg) as compared to CEV (34.95 ± 2.28 mg/mg). The level of MDA in the ovary of PS-MPs treated groups (P 2.5 mg/kg: 9.02 ± 1.19 ng/mg, P 5 mg/kg: 9.27 ± 1.41 ng/mg, P 10 mg/kg: 15.34 ± 1.87 ng/mg) was insignificantly different as compared to the normal control group. Moreover, the MDA level of PS-MP treated groups was statistically lower than of the ovarian tissue of CEV. The effect of PS-MP exposure on MDA in ovarian tissue of Wistar rats is shown in Fig. 2c.

SOD activity

In normal female rats, the SOD activity in the ovary of normal control group (8.57 ± 0.41 U/mg protein) was higher than the CEV (5.99 ± 0.58 U/mg protein). All PS-MP treated groups (P 2.5: 6.29 ± 0.73 U/mg protein, P 5 mg/kg: 5.69 ± 0.61 U/mg protein, P 10 mg/kg: 5.63 ± 0.59 U/mg protein) had considerably decreased activity of SOD in ovarian tissue than normal control group. The effect of PS-MP treatment on SOD activity in ovarian tissues is statistically comparable to CEV. The effect of PS-MP exposure on SOD activity in ovarian tissue of Wistar rats is shown in Fig. 2d.

Effect on lipid profile

The levels of TC, TG, and LDL in CEV were statistically similar to the normal control group. Treatment of female rats with PS-MP at all dose levels significantly raised TC (90.66 ± 3.02 mg/dl, P 5 mg/kg: 84.02 ± 2.94 mg/dl, P 10 mg/kg: 88.13 ± 4.88 mg/dl), TG (P 2.5 mg/kg: 45.66 ± 2.02 mg/dl, P 5 mg/kg: 57.18 ± 3.23 mg/dl, P 10 mg/kg: 58.33 ± 2.67 mg/dl), and LDL (P 2.5 mg/kg: 51.66 ± 2.33 mg/dl, P 5 mg/kg: 57.63 ± 4.08 mg/dl, P 10 mg/kg:

60.33 ± 4.39 mg/dl) in comparison to the normal control group (TC: 45.66 ± 2.18 mg/dl, TG: 38.16 ± 2.12 mg/dl, LDL: 14.12 ± 3.21 mg/dl) and CEV. In normal rats, the level of HDL (20.66 ± 1.64 mg/dl) was significantly lower than CEV (34.33 ± 2.62 mg/dl). Treatment with PS-MP 2.5 mg/kg resulted in significantly reduced level of HDL (17 ± 1.11 mg/dl) in comparison to the normal control group. Treatment with PS-MP at 5 (17.33 ± 1.24 mg/dl) and 10 mg/kg (18.1 ± 1.32 mg/dl) resulted in statistically insignificant alteration of HDL level as compared to the normal control group. The level of HDL in PS-MP treated groups was statistically lower than CEV. The effect of PS-MP exposure on lipid profile in female rats is shown in Fig. 3.

Influence on hormone levels

The hormone levels such as FSH, LH, testosterone, progesterone, and testosterone were determined 45 days post-exposure to PS-MP in female rats. In normal rats, the levels of FSH, LH, progesterone, and testosterone were significantly lower than CEV; however, the E2 level remained statistically indifferent among both groups. Administration of PS-MP at all doses raised the level of FSH (P 2.5 mg/kg: 137.73 ± 1.01 mIU/ml, P 5 mg/kg: 190.53 ± 1.8 mIU/ml, P 10 mg/kg: 207.26 ± 1.52 mIU/ml) and testosterone (P 2.5 mg/kg: 0.587 ± 0.03 ng/ml, P 5 mg/kg: 0.367 ± 0.04 ng/ml, P 10 mg/kg: 0.39 ± 0.01 ng/ml) as compared to the normal control group (FSH: 17.82 ± 2.02 mIU/ml, testosterone: 0.015 ± 0.004 ng/ml). The LH was significantly increased in estrogen-treated animals (12.04 ± 0.28 mIU/ml) in comparison to normal control rats (9.2 ± 0.27 mIU/ml). The level of LH was decreased at PS-MP 2.5 mg/kg (5.8 ± 0.27 mIU/ml), while progesterone was decreased at PS-MP 2.5 (116.41 ± 0.12 ng/ml) and 5 mg/kg (93.81 ± 0.09 ng/ml) as compared to normal control group (128.4 ± 0.15 ng/ml). At PS-MP 10 mg/kg, the level of E2 (3.09 ± 0.004 ng/ml) was statistically elevated as compared to normal control group (2.73 ± 0.005 ng/ml). The effect of PS-MP exposure on FSH, LH, E2, testosterone, and progesterone in female rats is shown in Fig. 4.

Effect on blood glucose

In normal rats, OGTT was unaffected in the CEV as compared to normal control group. In PS-MP 5 and 10 mg/kg group, the blood glucose concentration was significantly increased at 120 and 180 min as compared to normal control group. Moreover, in PS-MP 10 mg/kg, the blood glucose concentration was substantially higher at 0 min in comparison to normal control group. The effect of PS-MP exposure on OGTT in PS-MP exposed female rats is shown in Fig. 5.

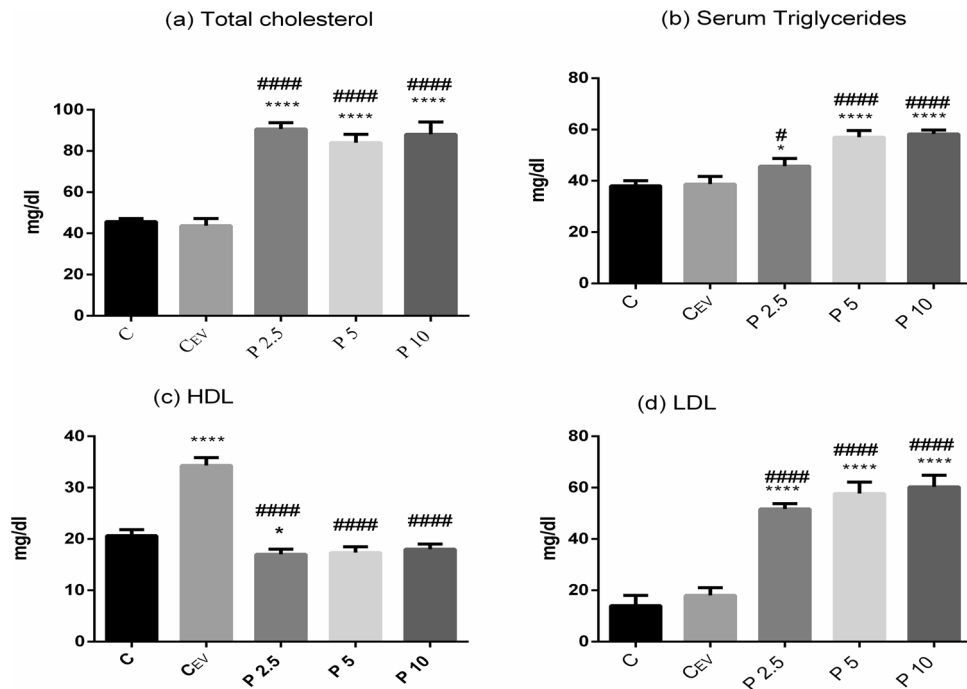


Fig. 3 Effect of polystyrene microplastic exposure on lipid biomarkers in female Wistar rats (a) Total cholesterol, (b) Serum triglycerides (c) High-density lipoprotein, and (d) Low-density lipoprotein in serum. Results were calculated as mean $\pm$ S.D ($n=6$). Data were assessed by using one-way ANOVA followed by Bonferroni's post hoc test. **** $p < 0.0001$ and * $p < 0.05$ showed considerably different in comparison to normal control group (C), whereas ##### $p < 0.0001$

and # $p < 0.05$ showed considerably different in comparison to estrogen-treated group (CEV). P 2.5: polystyrene microplastic 2.5 mg/kg/day, P 5: polystyrene microplastic 5 mg/kg/day, P 10: polystyrene microplastic 10 mg/kg/day. The exposure of PS-MP significantly increased total cholesterol, triglycerides, and low-density lipoprotein (LDL) in ovarian tissues. It also reduced high-density lipoprotein (HDL) at 2.5 mg/kg dose

Serum insulin

In female rats, serum insulin level in the CEV (0.97 ± 0.026 μ IU/ml) was considerably higher than in the normal control group (0.135 ± 0.018 μ IU/ml). When compared to the normal control group, all PS-MP treated groups (P 2.5 mg/kg: 0.55 ± 0.1 μ IU/ml, P 5 mg/kg: 0.96 ± 0.03 μ IU/ml, P 10 mg/kg: 0.86 ± 0.04 μ IU/ml) had shown substantially higher serum insulin level in comparison to the normal control group. It was also found that PS-MP 5 and 10 mg/kg groups demonstrated considerable increase in insulin level that was statistically comparable to CEV. Figure 5b shows the serum insulin findings in female rats on sub-chronic exposure to PS-MP.

Level of Nrf-2, IL-6, and NF- κ B

In normal rats, level of IL-6, NF- κ B, and Nrf-2 were increased in CEV (IL-6: 151.57 ± 27.8 pg/mL, NF- κ B: 109.24 ± 13.3 pg/mL, Nrf-2: 131.35 ± 3.17 pg/mL) than in the normal control group (IL-6: 45.54 ± 7.9 pg/mL, NF- κ B: 44.15 ± 5.59 pg/mL, Nrf-2: 54.15 ± 1.89 pg/mL). In all PS-MP treated groups, the level of Nrf-2 (P 2.5 mg/kg: 57.31 ± 1.57 pg/mL, P 5 mg/kg: 56.08 ± 3.08 pg/mL, P 10 mg/kg: 56.26 ± 1.05 pg/mL) remained comparable

to the normal control group. In PS-MP 2.5 and 5 mg/kg, the level of IL-6 (P 2.5 mg/kg 48.11 ± 4.7 pg/mL, P 5 mg/kg: 45.81 ± 1.1 pg/mL) and NF- κ B (P 2.5 mg/kg: 53.63 ± 3.36 pg/mL, P 5 mg/kg: 58.28 ± 2.77 pg/mL) remained insignificantly different as compared to the normal control group. However, in PS-MP 10 mg/kg group, the quantities of IL-6 (108.54 ± 17 pg/mL) and NF- κ B (103.54 ± 3.24 pg/mL) were significantly increased with respect to the normal control group. It was also demonstrated that the level of Nrf2, IL-6, and NF- κ B was significantly lower than CEV. The results of IL-6, NF- κ B, and Nrf-2 in the PS-MP exposed female rats are shown in Fig. 5.

Histopathology of liver

Histopathological slides of the liver of female rats treated with PS-MPs were examined by comparing it with the normal control group. In the normal control group, fibrosis or immune cell infiltration was not found. A slight amount of fibrosis was observed in female rats treated with PS-MPs; however, immune cell infiltration was not found. Activation of stellate cells was seen in PS-MP-treated rats. Hepatocytes of PS-MPs exposed rats hardly showed intracellular fat deposition as shown in Fig. 6.

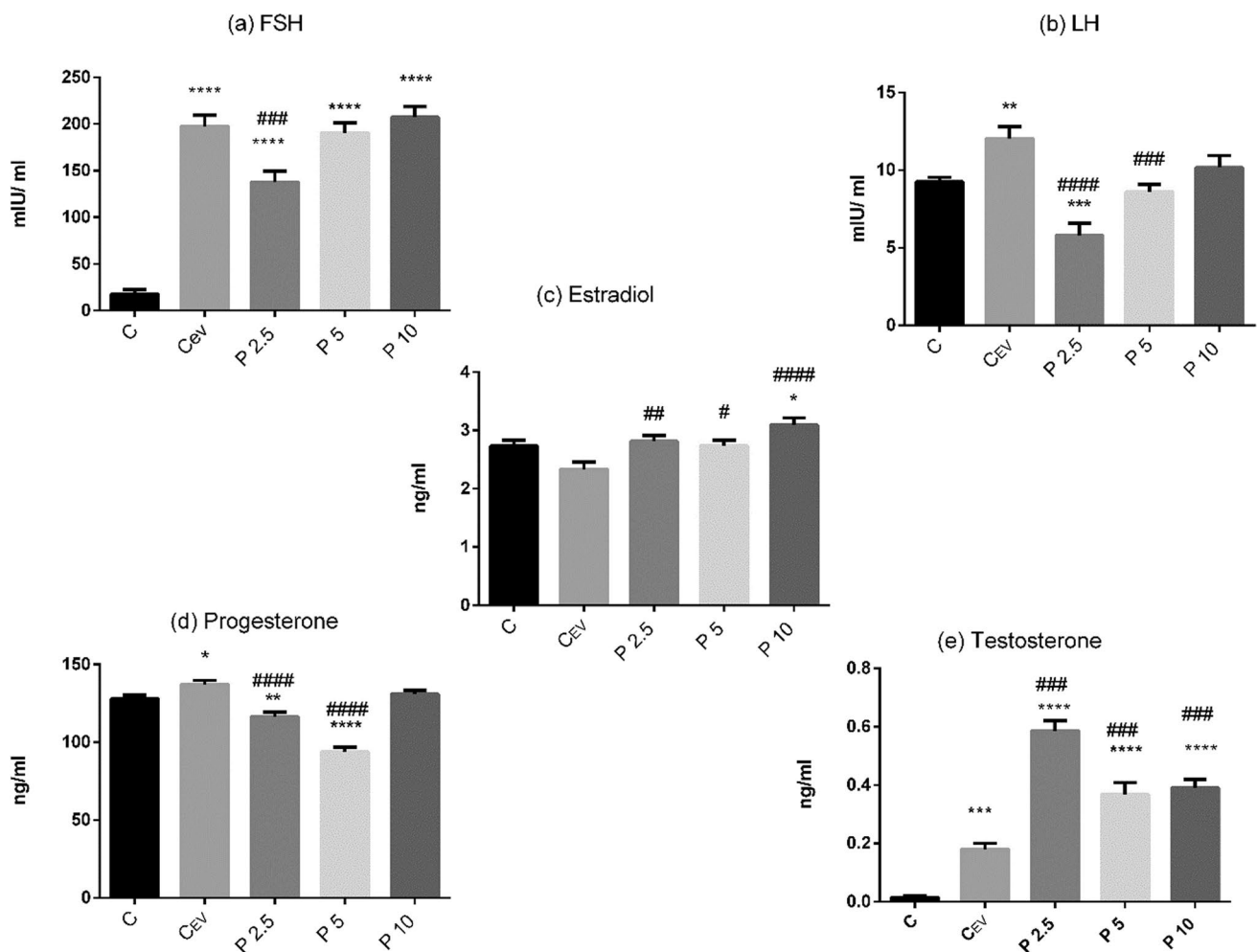


Fig. 4 Effect of polystyrene microplastic exposure on sex hormones in the serum of female rats. **(a)** Follicle stimulating hormone, **(b)** Leutinizing hormone, **(c)** Estradiol, **(d)** Progesterone, and **(e)** Testosterone in serum. Results were calculated as mean $\pm$ S.D ($n=6$). Data were assessed by using one-way ANOVA followed by Bonferroni's post hoc test. **** $p<0.0001$; *** $p<0.001$; ** $p<0.01$; * $p<0.05$ showed considerably different in comparison to normal control group (C), whereas ##### $p<0.0001$; ### $p<0.001$; ## $p<0.01$; # $p<0.05$

showed considerably different in comparison to estrogen-treated group (CEV). P 2.5: polystyrene microplastic 2.5 mg/kg/day, P 5: polystyrene microplastic 5 mg/kg/day, P 10: polystyrene microplastic 10 mg/kg/day. The exposure of PS-MPs significantly increased follicle stimulating hormone (FSH) and testosterone at all doses while increased estradiol was found only at 10 mg/kg dose. Progesterone and luteinizing hormone (LH) were reduced or unchanged on exposure to PS-MPs

Histopathology of ovary

The ovary of rats treated with PS-MPs was evaluated and compared to that of normal control. In the normal control group, no mature follicles were seen; however, primary follicles were seen and most area was covered with corpus luteum. In PS-MP exposed rats, the ovary seemed appropriately functional, primary and secondary follicles were present, and corpus luteum was found and intact as shown in Fig. 7.

Discussion

Previous studies have described that PS-MP-related infertility in male rats was associated with the alteration in the reproductive hormones (Amereh et al. 2020). It was found that the mitotic and meiotic divisions within the male reproductive system, DNA damage, and the decline in sperm motility were attributed to ROS generation as well as altered paracrine and endocrine signaling (Jin et al. 2021). PS-MPs are more accumulated in ovaries as compared to testes which

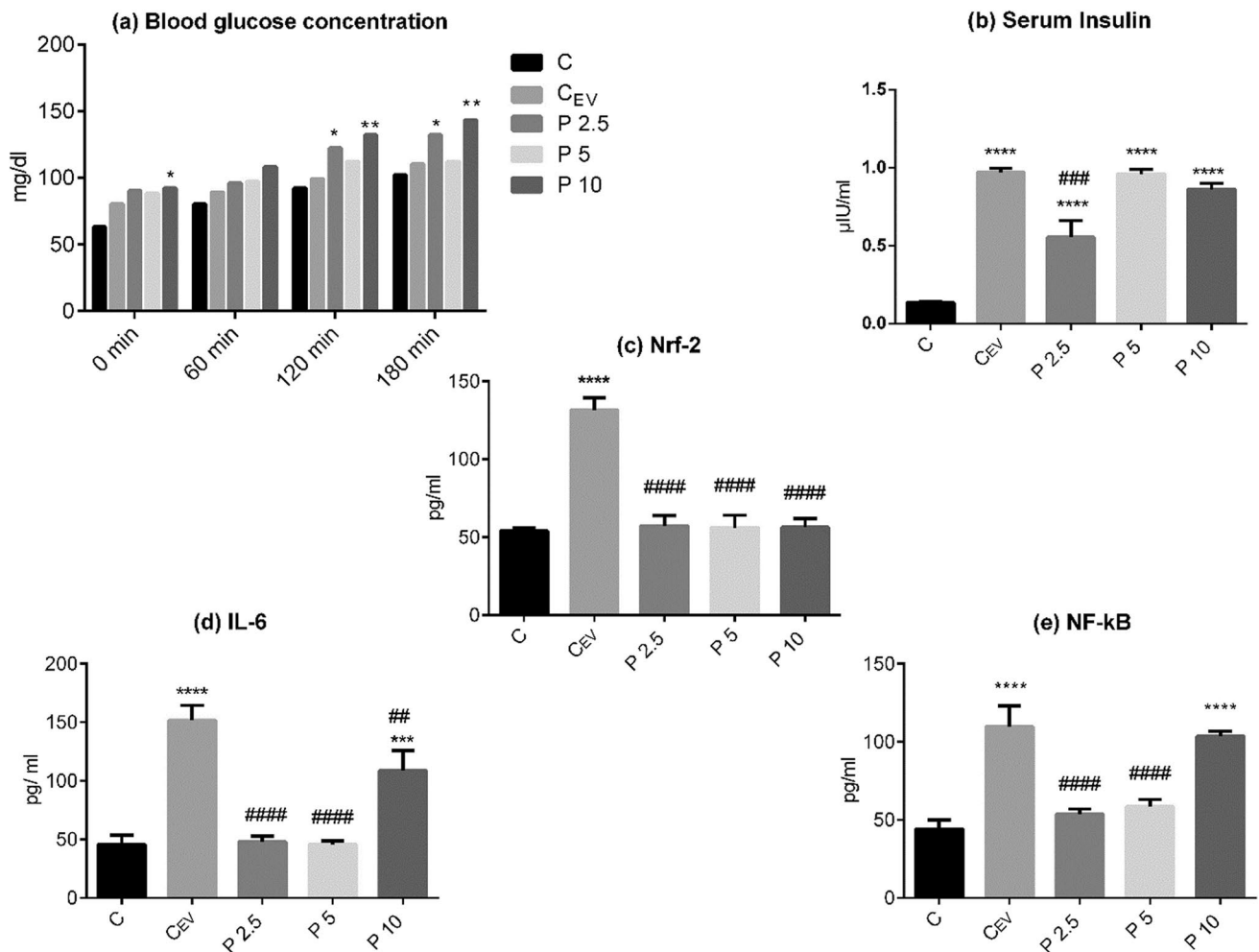


Fig. 5 Effect of polystyrene microplastic exposure on blood glucose, insulin, and other biomarkers in the serum of female rats. **(a)** Blood glucose level, **(b)** serum insulin level, **(c)** Nrf-2 level, **(d)** IL-6 level, and **(e)** NF-κB level in serum. Results were calculated as mean $\pm$ S.D ($n=6$). Data were assessed by using one-way ANOVA followed by Bonferroni's post-hoc test. **** $p < 0.0001$; *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$ showed considerably different in comparison to normal control group (C), whereas #### $p < 0.0001$; ### $p < 0.001$;

$p < 0.01$ showed considerably different in comparison to estrogen-treated group (CEV). P 2.5: polystyrene microplastic 2.5 mg/kg/day, P 5: polystyrene microplastic 5 mg/kg/day, P 10: polystyrene microplastic 10 mg/kg/day. PS-MPs resulted in insulin resistance as evident from the increased insulin and high blood glucose level in oral glucose tolerance test. The PS-MP increased serum IL-6 and NF-κB at 10 mg/kg dose; however, Nrf2 remained unaffected

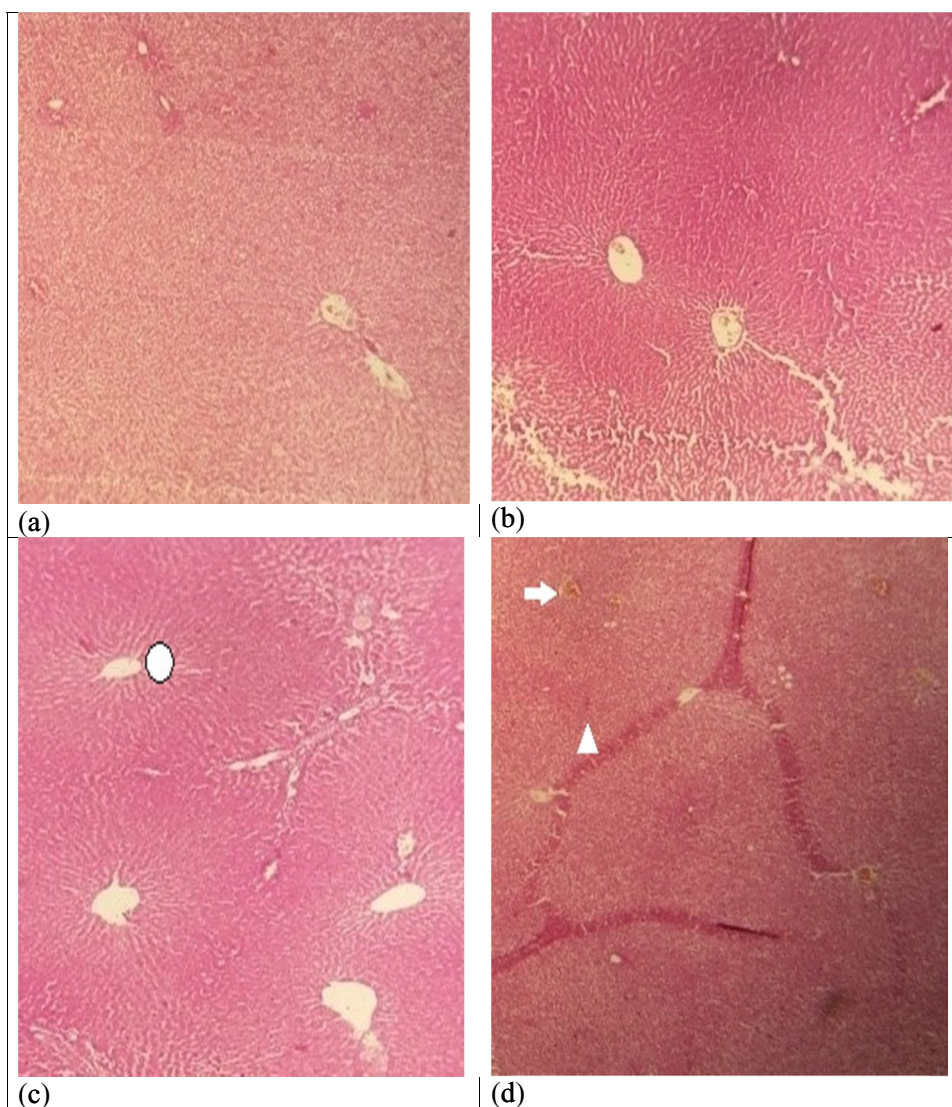
make the female reproductive system more prone to toxic effects (Wei et al. 2022). Therefore, the current research demonstrated the consequences of PS-MPs exposure on female rats. The effects of PS-MPs exposure at different dose levels on lipid profile, serum insulin, liver function, reproductive hormones, and histopathology of the liver and ovaries were observed.

Current research showed the development of oxidative stress as one of the key mechanisms of PS-MP-induced damage in female rats. It was found that PS-MPs reduced the activity of CAT and SOD in the liver of Wistar rats with only moderate elevation of MDA. SOD and CAT are antioxidant enzymes that convert superoxide anions (O_2^-) into H_2O_2 , and then H_2O_2 into H_2O and O_2 respectively.

MDA, a hazardous end-product of lipid peroxidation, represents lipid peroxidation. GSH assists the conversion of peroxides into less-toxic hydroxyl molecules which can effectively reduce the formation of ROS (Kirici et al. 2017). In a previous study, it was found that oxidative stress biomarkers such as SOD and CAT were considerably elevated in zebrafish liver treated with 5-μm PS-MPs (Lu et al. 2016). PS-MP ($< 1 \mu m$) exposure was also correlated to oxidative stress, impaired immunity, and diminished energy metabolism at the cellular and molecular level in rat liver (Avio et al. 2015).

This study showed the induction of oxidative stress in the ovarian tissue of Wistar rats. Previously, it is found that MPs slowed ovarian development, reduced fecundity,

Fig. 6 Histopathology of liver in polystyrene microplastic (PS-MP)-treated female rats at 10× magnification. **a** Normal control group. **b** PS-MP 2.5 mg/kg/day treated group. **c** PS-MP 5 mg/kg/day treated group. **d** PS-MP 10 mg/kg/day treated group. Circle shows normal central vein. Arrow shows congested portal vein. Triangle shows fibrosis



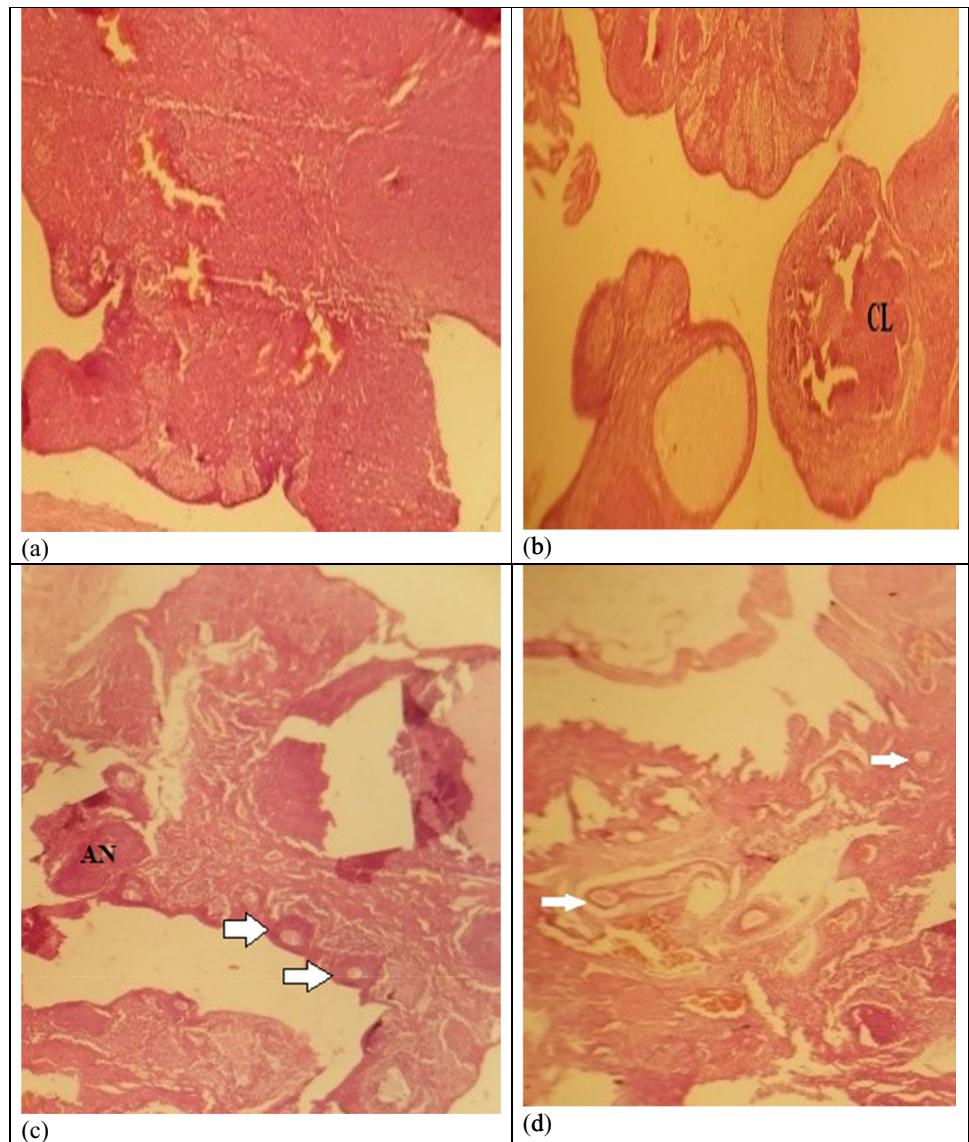
and disrupted reproduction in marine medaka (*Oryzias melastigma*). The PS-MPs of 0.5 μm size were detected in ovaries of the *Carcinus maenas* (An et al. 2021). Another signaling pathway such as Wnt/ β -catenin could be triggered by oxidative stress. Moreover, increase in ROS could cause granulosa cell's death resulting in follicular atresia (Shen et al. 2012). Therefore, it can be speculated that oxidative stress induced by PS-MPs could trigger endocrine disruption in female rats.

According to earlier investigations, the serum insulin was higher in the animals suffering from polycystic ovary syndrome (PCOS) than the normal. Insulin resistance is thought to be pivotal in developing PCOS and documented in 25 to 60% of women afflicted with the disease (Yeon Lee et al. 2010). In the present study, the serum level of insulin was elevated in animals treated with PS-MPs along with glucose intolerance, which predisposes the animals

to PCOS (Abtahi-Eivari et al. 2018). Moreover, FSH and LH are increased in the PCOS (Abtahi-Eivari et al. 2018). In a prior study on PCOS rats, a higher amount of LH, FSH, testosterone, and E2 were evident (Abtahi-Eivari et al. 2018). Similar results were seen in the current study which showed endocrine disruption caused by PS-MPs in female rats (Amereh et al. 2020).

The liver, a main metabolic organ, alters lipid metabolism in response to environmental toxicants (Van den Berghe 1991). After exposure to MPs, the content of triglycerides was significantly higher in the liver of zebrafish *Danio rerio* (Chen et al. 2017). The current study showed a rise in TC, TG, and LDL as well as decline in HDL in the PS-MPs exposed rats which demonstrated the metabolic dysfunction and increased risk of cardiovascular events on sub-chronic exposure to PS-MPs. The damage to liver was evident from fibrosis and activation of stellate cells. This

Fig. 7 Histopathology of ovary in polystyrene microplastic (PS-MP)-treated female rats at 10× magnification. **a** Normal control group. **b** PS-MP 2.5 mg/kg/day treated group. **c** PS-MP 5 mg/kg/day treated group. **d** PS-MP 10 mg/kg/day treated group. AN antral follicle, CL corpus luteum, arrows show developing follicles



alteration of lipid profile may be attributed to interaction of MPs with lipase and lipid droplets to limit lipid utilization by blocking the activated protein kinase-peroxisome proliferator-activated receptor- α and downregulating the lipolysis-related genes (Tan et al. 2020).

There was significantly increased level of serum IL-6 and NF- κ B in PS-MP-exposed rats. According to a previous investigation, the expression of IL-6 in PCOS rats was increased which controlled cell proliferation, cell survival, differentiation, immune cell activation, cell migration, and cell death (Ghowasi et al. 2018);(Caglayan et al. 2018). The IL-6 functions as a defense mechanism, but is proinflammatory during chronic inflammation (Masjedi et al. 2018). The NF- κ B is an important regulator of the hosts' inflammatory and immunological responses that prevents the cell death in

response to cellular stressors (Balci et al. 2020);(Lingappan 2018). Elevation of IL-6 and NF- κ B in PS-MPs exposed rats suggests the cellular damage through chronic inflammation and early apoptosis (Mohammed et al. 2020);(Bilal et al. 2019).

An earlier study showed that the 0.5- μ m-sized PS-MPs could enter into the ovarian granulosa cells of rats and induce the oxidative stress resulting in granulosa cell apoptosis. PS-MPs also activate the Wnt/ β -catenin signaling pathway that caused ovarian fibrosis (An et al. 2021). In the current study, the ovaries of rats exposed to PS-MPs seemed appropriately functional which might be due to increased size of PS-MPs. There are several limitations of current study including the lack of gene expression of liver and ovarian tissues, which would

demonstrate the mechanism behind the metabolic and reproductive changes in female rats induced by PS-MPs. The staining of early apoptotic markers, such as caspases as well as for ROS-induced changes, in the liver and ovary must also be carried out to visualize the oxidative stress-related damage in these tissues.

Conclusion

The level of CAT and SOD in the liver and ovarian tissues of rats was significantly altered by PS-MPs. The exposure to PS-MPs exhibited an increase in the levels of TG, TC, LDL, E2, testosterone, FSH, and insulin in female rats, while decrease in the levels of HDL was evident in female rats. PS-MP exposure also caused an elevation of IL-6 and NF- κ B in serum of exposed female rats. Histopathology of the liver showed slight fibrosis and activation of stellate cells. These findings suggest that the sub-chronic exposure to PS-MPs in female rats caused the metabolic and endocrine dysfunctions due to hormonal imbalance and induction of oxidative stress, IL-6, and NF- κ B.

Author contribution MFA contributed to the study's conception and design. A. Saeed, A. Saleem, BA and A. Sharif performed the experiment and data analysis. MFA and A. Saleem wrote the first draft of the manuscript. All the authors approved the final manuscript.

Data availability All data are available with the author and will be made available on reasonable request.

Declarations

Ethical approval The study was certified by the Ethics Committee of Riphah International University, Lahore Reference No. REC/RIPS-LHR/2022/518. Furthermore, to minimize animal suffering, National Institute of Health (NIH) guidelines were followed for the experimental methodologies.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

References

Abtahi-Eivari SH, Moghimi M, Soltani M, Shoorei H, Asghari R, Hajizadeh H, Shokoohi M, Alami S, Ghaderi FK (2018) The effect of Galega officinalis on hormonal and metabolic profile in a rat model of polycystic ovary syndrome. *Intl J Womens Health Reprod Sci* 6:276–282

Akhtar MF, Younas S, Saleem A, Baig MMFA, Sharif A, Abdel-Daim MM, Rasul A, Saleem M (2021a) Maternotoxicity and fetotoxicity in *Rattus norvegicus albinus* exposed to tramadol during the late phase of pregnancy. *Birth Defects Res* 113:1407–1421

Akhtar MF, Zubair S, Saleem A, Alsharif KF, Abdel-Daim MM (2021b) Comparison of individual and combination treatments with naproxen, prednisolone and hydroxychloroquine to treat complete Freund's adjuvant induced arthritis. *Inflammopharmacology* 29:1719–1731

Amereh F, Babaei M, Eslami A, Fazelipour S, Rafiee M (2020) The emerging risk of exposure to nano (micro) plastics on endocrine disturbance and reproductive toxicity: from a hypothetical scenario to a global public health challenge. *Environ Pollut* 261:114158

Amini L, Ghorbani B, Afshar B (2020) The comparison of infertility stress and perceived social support in infertile women and spouses of infertile men. *Iran J Nurs* 32:74–85

An R, Wang X, Yang L, Zhang J, Wang N, Xu F, Hou Y, Zhang H, Zhang L (2021) Polystyrene microplastics cause granulosa cells apoptosis and fibrosis in ovary through oxidative stress in rats. *Toxicology* 449:152665

Astner A, Hayes D, O'Neill H, Evans B, Pingali S, Urban V, Young T (2019) Mechanical formation of micro-and nano-plastic materials for environmental studies in agricultural ecosystems. *Sci Total Environ* 685:1097–1106

Avio CG, Gorbi S, Milan M, Benedetti M, Fattorini D, d'Errico G, Paoletto M, Bargelloni L, Regoli F (2015) Pollutants bioavailability and toxicological risk from microplastics to marine mussels. *Environ Pollut* 198:211–222

Balci A, Ozkemahli G, Erkekoglu P, Zeybek ND, Yersal N, Kocer-Gumusel B (2020) Histopathologic, apoptotic and autophagic, effects of prenatal bisphenol A and/or di(2-ethylhexyl) phthalate exposure on prepubertal rat testis. *Environ Sci Pollut Res* 27:20104–20116

Bhangale JO, Acharya SR (2016): Anti-Parkinson activity of petroleum ether extract of *Ficus religiosa* (L.) leaves. *Advances in pharmacological sciences* 2016

Bilal M, Iqbal HMN, Barceló D (2019) Mitigation of bisphenol A using an array of laccase-based robust bio-catalytic cues – a review. *Sci Total Environ* 689:160–177

Bin-Jumah MN, Al-Huqail AA, Abdelnaeem N, Kamel M, Fouda MMA, Abulmeaty MMA, Saadeldin IM, Abdel-Daim MM (2021) Potential protective effects of *Spirulina platensis* on liver, kidney, and brain acrylamide toxicity in rats. *Environ Sci Pollut Res* 28:26653–26663

Browne MA, Crump P, Niven SJ, Teuten E, Tonkin A, Galloway T, Thompson R (2011) Accumulation of microplastic on shorelines worldwide: sources and sinks. *Environ Sci Technol* 45:9175–9179

Caglayan C, Kandemir FM, Yıldırım S, Kucukler S, Kılınc MA, Saglam YS (2018) Zingerone ameliorates cisplatin-induced ovarian and uterine toxicity via suppression of sex hormone imbalances, oxidative stress, inflammation and apoptosis in female wistar rats. *Biomed Pharmacother* 102:517–530

Capdevila S, Giral M, Ruiz de La Torre J, Russell R, Kramer K (2007) Acclimatization of rats after ground transportation to a new animal facility. *Lab Anim* 41:255–261

Chen Q, Gundlach M, Yang S, Jiang J, Velki M, Yin D, Hollert H (2017) Quantitative investigation of the mechanisms of microplastics and nanoplastics toward zebrafish larvae locomotor activity. *Sci Total Environ* 584:1022–1031

Chiu H-W, Xia T, Lee Y-H, Chen C-W, Tsai J-C, Wang Y-J (2015) Cationic polystyrene nanospheres induce autophagic cell death through the induction of endoplasmic reticulum stress. *Nanoscale* 7:736–746

Cole M, Lindeque P, Halsband C, Galloway TS (2011) Microplastics as contaminants in the marine environment: a review. *Mar Pollut Bull* 62:2588–2597

Cox KD, Covernton GA, Davies HL, Dower JF, Juanes F, Dudas SE (2019) Human consumption of microplastics. *Environ Sci Technol* 53:7068–7074

Crawford C, Quinn B (2017): The interactions of microplastics and chemical pollutants. *Microplastic pollutants*, 131–157

- Deshpande PS, Gupta AS (2019) Causes and prevalence of factors causing infertility in a public health facility. *J Human Reprod Sci* 12:287
- Ensign LM, Cone R, Hanes J (2012) Oral drug delivery with polymeric nanoparticles: the gastrointestinal mucus barriers. *Adv Drug Deliv Rev* 64:557–570
- Eriksen M, Lebreton LC, Carson HS, Thiel M, Moore CJ, Borerro JC, Galgani F, Ryan PG, Reisser J (2014) Plastic pollution in the world's oceans: more than 5 trillion plastic pieces weighing over 250,000 tons afloat at sea. *PLoS ONE* 9:e111913
- Forte M, Iachetta G, Tussellino M, Carotenuto R, Prisco M, De Falco M, Laforgia V, Valiante S (2016) Polystyrene nanoparticles internalization in human gastric adenocarcinoma cells. *Toxicol in Vitro* 31:126–136
- Fried JR (2014): Polymer science and technology. Pearson Education
- Geiser M, Rothen-Rutishauser B, Kapp N, Schürch S, Kreyling W, Schulz H, Semmler M, Hof VI, Heyder J, Gehr P (2005) Ultrafine particles cross cellular membranes by nonphagocytic mechanisms in lungs and in cultured cells. *Environ Health Perspect* 113:1555–1560
- Germoush MO, Fouda MMA, Kamel M, Abdel-Daim MM (2022) *Spirulina platensis* protects against microcystin-LR-induced toxicity in rats. *Environ Sci Pollut Res* 29:11320–11331
- Ghowri M, Khazali H, Sisakhtnezhad S (2018) Evaluation of TNF- α and IL-6 mRNAs expressions in visceral and subcutaneous adipose tissues of polycystic ovarian rats and effects of resveratrol. *Iran J Basic Med Sci* 21:165
- Guzzetti E, Sureda A, Tejada S, Faggio C (2018) Microplastic in marine organism: environmental and toxicological effects. *Environ Toxicol Pharmacol* 64:164–171
- Hussain MA, Abogresha NM, Hassan R, Tamany DA, Lotfy M (2016) Effect of feeding a high-fat diet independently of caloric intake on reproductive function in diet-induced obese female rats. *Arch Med Sci: AMS* 12:906
- Iftikhar A, Akhtar MF, Saleem A, Riaz A, Zehravi M, Rahman M, Md Ashraf G Comparative potential of zinc sulfate, l-carnitine, lycopene, and coenzyme Q10 on cadmium-induced male infertility. *Intl J Endocrinol* 2022, 1–11
- Ijaz MU, Shahzadi S, Samad A, Ehsan N, Ahmed H, Tahir A, Rehman H, Anwar H (2021) Dose- dependent effect of polystyrene microplastics on the testicular tissues of the male Sprague Dawley Rats. *Dose-Response* 19:15593258211019882
- Jin H, Ma T, Sha X, Liu Z, Zhou Y, Meng X, Chen Y, Han X, Ding J (2021) Polystyrene microplastics induced male reproductive toxicity in mice. *J Hazard Mater* 401:123430
- Kabir MT, Rahman MH, Akter R, Behl T, Kaushik D, Mittal V, Pandey P, Akhtar MF, Saleem A, Albadrani GM, Kamel M, Khalifa SAM, El-Seedi HR, Abdel-Daim MM (2021) Potential role of curcumin and its nanoformulations to treat various types of cancers. *Biomolecules* 11:392
- Kirici M, Turk C, Caglayan C, Kirici M (2017) Toxic effects of copper sulphate pentahydrate on antioxidant enzyme activities and lipid peroxidation of freshwater fish *Capoeta umbla* (Heckel, 1843) tissues. *Appl Ecol Environ Res* 15:1685–1696
- Kosuth M, Mason SA, Wattenberg EV (2018) Anthropogenic contamination of tap water, beer, and sea salt. *PLoS ONE* 13:e0194970
- Kumar MV, Gupta Y (2002) Effect of different extracts of *Centella asiatica* on cognition and markers of oxidative stress in rats. *J Ethnopharmacol* 79:253–260
- Lehner R, Weder C, Petri-Fink A, Rothen-Rutishauser B (2019) Emergence of nanoplastic in the environment and possible impact on human health. *Environ Sci Technol* 53:1748–1765
- Li J, Qu X, Su L, Zhang W, Yang D, Kolandhasamy P, Li D, Shi H (2016) Microplastics in mussels along the coastal waters of China. *Environ Pollut* 214:177–184
- Lingappan K (2018) NF- κ B in oxidative stress. *Curr Opin Toxicol* 7:81–86
- Lodge TP (2017): Celebrating 50 years of macromolecules. ACS Publications
- Lokind KB, Lorenzen FH, Bundgaard H (1991) Oral bioavailability of 17 β -estradiol and various ester prodrugs in the rat. *Int J Pharm* 76:177–182
- Lu Y, Zhang Y, Deng Y, Jiang W, Zhao Y, Geng J, Ding L, Ren H (2016) Uptake and accumulation of polystyrene microplastics in zebrafish (*Danio rerio*) and toxic effects in liver. *Environ Sci Technol* 50:4054–4060
- Manikkam M, Tracey R, Guerrero-Bosagna C, Skinner MK (2013) Plastics derived endocrine disruptors (BPA, DEHP and DBP) induce epigenetic transgenerational inheritance of obesity, reproductive disease and sperm epimutations. *PLoS ONE* 8:e55387
- Mansouri E, Panahi M, Ghaffari MA, Ghorbani A (2011) Effects of grape seed proanthocyanidin extract on oxidative stress induced by diabetes in rat kidney. *Iran Biomed J* 15:100
- Masjedi A, Hashemi V, Hojjat-Farsangi M, Ghalamfarsa G, Azizi G, Yousefi M, Jadidi-Niaragh F (2018) The significant role of interleukin-6 and its signaling pathway in the immunopathogenesis and treatment of breast cancer. *Biomed Pharmacother* 108:1415–1424
- Mirabolghasemi G, Kamyab Z (2017) Changes of the uterine tissue in rats with polycystic ovary syndrome induced by estradiol valerate. *Intl J Fertil Steril* 11:47
- Mohammed ET, Hashem KS, Ahmed AE, Aly MT, Aleya L, Abdel-Daim MM (2020) Ginger extract ameliorates bisphenol A (BPA)-induced disruption in thyroid hormones synthesis and metabolism: involvement of Nrf-2/HO-1 pathway. *Sci Total Environ* 703:134664
- Park E-J, Han J-S, Park E-J, Seong E, Lee G-H, Kim D-W, Son H-Y, Han H-Y, Lee B-S (2020) Repeated-oral dose toxicity of polyethylene microplastics and the possible implications on reproduction and development of the next generation. *Toxicol Lett* 324:75–85
- Piringer OG, Baner AL (2008) Plastic packaging materials for food: barrier function, mass transport, quality assurance, and legislation. John Wiley & Sons
- Pitt JA, Kozal JS, Jayasundara N, Massarsky A, Trevisan R, Geitner N, Wiesner M, Levin ED, Di Giulio RT (2018) Uptake, tissue distribution, and toxicity of polystyrene nanoparticles in developing zebrafish (*Danio rerio*). *Aquat Toxicol* 194:185–194
- Prata JC, da Costa JP, Lopes I, Duarte AC, Rocha-Santos T (2020) Environmental exposure to microplastics: an overview on possible human health effects. *Sci Total Environ* 702:134455
- Rossi G, Barnoud J, Monticelli L (2014) Polystyrene nanoparticles perturb lipid membranes. *J Phys Chem Lett* 5:241–246
- Saleem A, Akhtar MF, Latif A, Abdel-Daim MM, Baig MMFA, Saleem M, Mehmood MH (2021): Chemical characterisation and appraisal of antidiabetic potential of *Terminalia citrina* extract in streptozotocin induced hyperglycaemia in Wistar rats. *Arch Physiol Biochem*, 1–14
- Saleem A, Khalid H, Akhtar MF, Zeb A (2022) HPLC-DAD analysis, anti-inflammatory and anti-arthritis potentials of *Coronopus didymus* (L.) Sm. extracts: effects on pro-and anti-inflammatory cytokines, COX-2, I- κ B, NF- κ B and oxidative stress biomarkers. *Food Funct* 13:6244–6258
- Sharif Z, Akhtar MF, Sharif H, Saleem A, Khan MI, Riaz A (2022) Endocrine disruption: reproductive toxicity of glyceryl trinitrate and isosorbide mononitrate in male Wistar rats. *Andrologia* 54:e14482
- Shen M, Lin F, Zhang J, Tang Y, Chen W-K, Liu H (2012) Involvement of the up-regulated FoxO1 expression in follicular granulosa cell apoptosis induced by oxidative stress. *J Biol Chem* 287:25727–25740
- Tan H, Yue T, Xu Y, Zhao J, Xing B (2020) Microplastics reduce lipid digestion in simulated human gastrointestinal system. *Environ Sci Technol* 54:12285–12294
- Van den Berghe G (1991) The role of the liver in metabolic homeostasis: implications for inborn errors of metabolism. *J Inherit Metab Dis* 14:407–420
- Velma V, Tchounwou PB (2010) Chromium-induced biochemical, genotoxic and histopathologic effects in liver and kidney of goldfish, *Carassius auratus*. *Mutation Research/genetic Toxicology and Environmental Mutagenesis* 698:43–51

- Wang T, Bai J, Jiang X, Nienhaus GU (2012) Cellular uptake of nanoparticles by membrane penetration: a study combining confocal microscopy with FTIR spectroelectrochemistry. *ACS Nano* 6:1251–1259
- Wei Z, Wang Y, Wang S, Xie J, Han Q, Chen M (2022) Comparing the effects of polystyrene microplastics exposure on reproduction and fertility in male and female mice. *Toxicology* 465:153059
- West-Eberhard MJ (2019) Nutrition, the visceral immune system, and the evolutionary origins of pathogenic obesity. *Proc Natl Acad Sci* 116:723–731
- Yacobi NR, DeMaio L, Xie J, Hamm-Alvarez SF, Borok Z, Kim K-J, Crandall ED (2008) Polystyrene nanoparticle trafficking across alveolar epithelium. *Nanomedicine: Nanotechnology, Biology and Medicine* 4:139–145
- Yeon Lee J, Baw C-K, Gupta S, Aziz N, Agarwal A (2010) Role of oxidative stress in polycystic ovary syndrome. *Curr Women's Health Rev* 6:96–107
- Zulfqar F, Akhtar MF, Saleem A, Akhtar B, Sharif A, Saleem U (2020) Chemical characterization, antioxidant evaluation, and antidiabetic potential of *Pinus gerardiana* (Pine nuts) extracts. *J Food Biochem* 44:e13199

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